

Aluffi Chapter-1 Solutions

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Exercise 0.1. *Alu09, Ex-1.6 Define a relation $\sim$ on $\mathbb{R}$ as $a \sim b$ if $a - b \in \mathbb{Z}$. Show that $\sim$ is an equivalence relation on $\mathbb{R}$. Find a compelling description for $\mathbb{R}/\sim$. Do the same for the relation $\equiv$ on the plane $\mathbb{R} \times \mathbb{R}$ defined by $(a_1, a_2) \equiv (b_1, b_2) \Leftrightarrow b_1 - a_1 \in \mathbb{Z}$ and $b_2 - a_2 \in \mathbb{Z}$.*

Proof. Reflexivity holds because $x - x = 0 \in \mathbb{Z}$. Symmetry is due to $y - x = -(x - y) \in \mathbb{Z}$. Finally, for $x, y, z \in \mathbb{R}$ with $x - y = a \in \mathbb{Z}$ and $y - z = b \in \mathbb{Z}$, we can write $x - z = x - y - (z - y) = a + b \in \mathbb{Z}$ which establishes transitivity. This completes the proof that $\sim$ is an equivalence relation.

Next, we will show that $\mathbb{R}/\sim$ is in bijection to the following set:

$$S^1 := \{x \in \mathbb{C}; \|x\| = 1\}$$

For this we introduce a map

$$g : \mathbb{R} \rightarrow S^1$$

defined by

$$x \mapsto e^{2\pi i x}$$

That g is well defined is obvious from intro analysis. We know from intro analysis that g is a surjection.

Then, we shall invoke *Alu09*, Thm-2.7 on the diagram

$$\begin{array}{ccc} \mathbb{R} & \xrightarrow{g} & S^1 \\ p \downarrow & \nearrow f & \\ \mathbb{R}/\sim & & \end{array}$$

This gives us a well defined injection $f : \mathbb{R}/\sim \rightarrow S^1$. We get that $\text{im } f = \text{im } g$ and since g is surjective $\text{im } f = S^1$ which completes the proof.

Next, we make an important observation.

Observe. *Since any pair of points in any set in $\mathbb{R}/\sim$ have the same fractional part, quotienting $\mathbb{R}$ by $\sim$ has the effect of treating the integral part of any real as insignificant and of course, those reals with identical fractional parts are identified as equivalent. Motivated by this, hereafter, we will simply use the notation $\mathbb{R}/\mathbb{Z}$ for $\mathbb{R}/\sim$.*

Since $\mathbb{R}/\mathbb{Z} \cong S^1$ we are tempted to believe that $\mathbb{R}^2/\mathbb{Z}^2$ is in bijection with $S^1 \times S^1$. We shall prove this.

First, notice that there is a bijection $f : \mathbb{R}/\mathbb{Z} \rightarrow S^1$. This gives us a bijection

$$\mathbb{R}/\mathbb{Z} \times \mathbb{R}/\mathbb{Z} \rightarrow S^1 \times S^1$$

by the rule

$$(x + \mathbb{Z}, y + \mathbb{Z}) \mapsto (f(x), f(y))$$

So, we simply need to prove that there is a bijection between $\mathbb{R}^2/\mathbb{Z}^2$ and $\mathbb{R}/\mathbb{Z} \times \mathbb{R}/\mathbb{Z}$.

Here, we shall use the universal property of products as in [Kim18, Prop-1.3.1](#).
For this, consider the function

$$f : \mathbb{R}^2/\mathbb{Z}^2 \rightarrow \mathbb{R}/\mathbb{Z}$$

defined by

$$(x, y) + \mathbb{Z} \times \mathbb{Z} \mapsto x + \mathbb{Z}$$

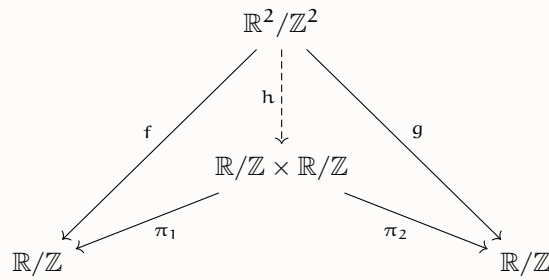
and the function

$$g : \mathbb{R}^2/\mathbb{Z}^2 \rightarrow \mathbb{R}/\mathbb{Z}$$

defined by

$$(x, y) + \mathbb{Z} \times \mathbb{Z} \mapsto y + \mathbb{Z}$$

Next, consider the diagram



By universal properties of products, there is a map

$$h : \mathbb{R}^2/\mathbb{Z}^2 \rightarrow \mathbb{R}/\mathbb{Z} \times \mathbb{R}/\mathbb{Z}$$

which satisfies

$$f = \pi_1 \circ h, \quad g = \pi_2 \circ h$$

It is obvious that f and g are surjective and from this follows the surjectivity of h .
Next, take any

$$(x, y) + \mathbb{Z}^2 \in \mathbb{R}^2/\mathbb{Z}^2$$

and

$$(x', y') + \mathbb{Z}^2 \in \mathbb{R}^2/\mathbb{Z}^2$$

so that $h((x, y) + \mathbb{Z}^2) = h((x', y') + \mathbb{Z}^2)$

Then, $(\pi_1 \circ h)((x, y) + \mathbb{Z}^2) = x + \mathbb{Z}$ and $(\pi_2 \circ h)((x, y) + \mathbb{Z}^2) = y + \mathbb{Z}$ and a similar result holds for $(x', y') + \mathbb{Z}^2$.

So, if

$$(x + \mathbb{Z}, y + \mathbb{Z}) = (x' + \mathbb{Z}, y' + \mathbb{Z})$$

then we have

$$x + \mathbb{Z} = x' + \mathbb{Z}, \quad y + \mathbb{Z} = y' + \mathbb{Z}$$

This tells us that $x \equiv x' \pmod{\mathbb{Z}}$ and $y \equiv y' \pmod{\mathbb{Z}}$. This is precisely the condition for

$$(x, y) \equiv (x', y') \pmod{\mathbb{Z}^2}$$

which is another way of saying that $(x, y) + \mathbb{Z}^2 = (x', y') + \mathbb{Z}^2$ which proves that h is an injection thereby completing the proof. $\square$

Remark. It is worth mentioning that the second half of 0.1 could be proved by invoking the universal property of the quotient $\mathbb{R}^2/\mathbb{Z}^2$ instead of invoking the universal property of the product $\mathbb{R}/\mathbb{Z} \times \mathbb{R}/\mathbb{Z}$.

References

- [Alu09] Paolo Aluffi. *Algebra: Chapter 0*. Vol. 104. Graduate Studies in Mathematics. American Mathematical Society, 2009. 713 pp. ISBN: 978-1-4704-6571-1. DOI: <https://doi.org/10.1090/gsm/104>. URL: <https://bookstore.ams.org/GSM/104>.
- [Kim18] Dongryul Kim. *A rough guide to linear algebra*. 2018. 151 pp. URL: <https://web.stanford.edu/~dkim04/writings/LinAlg.pdf>.